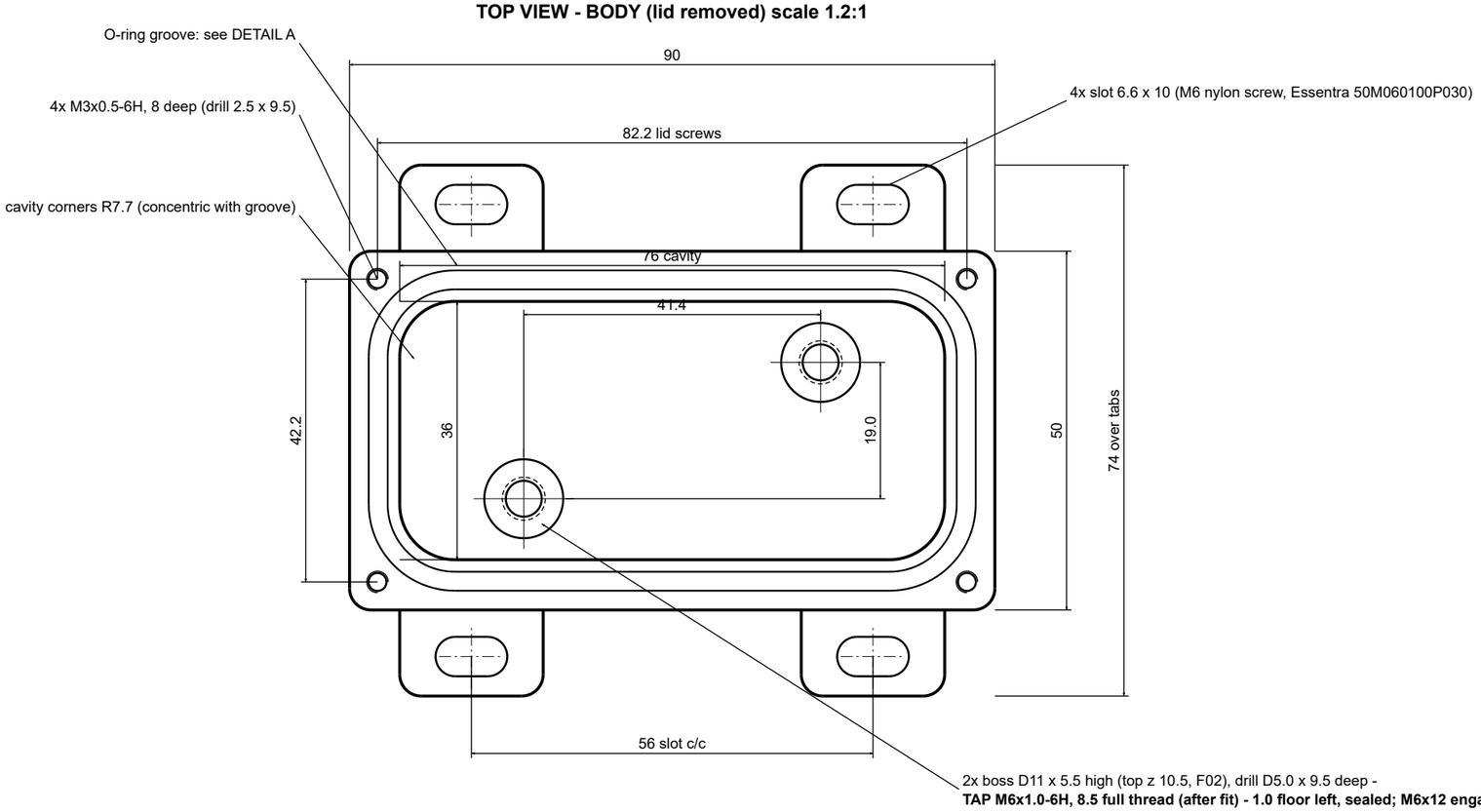
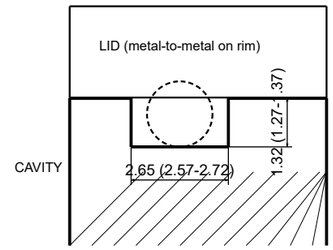
TOP VIEW - BODY (lid removed) scale 1.2:1



DETAIL A - O-ring groove (body rim) scale 6:1

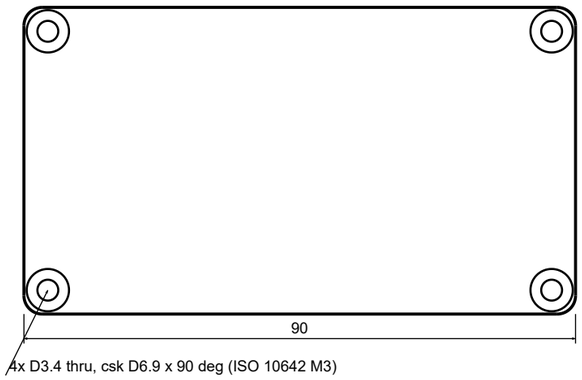


Parker ORD 5700 Design Chart 4-3, W 1.78 face seal, liquids

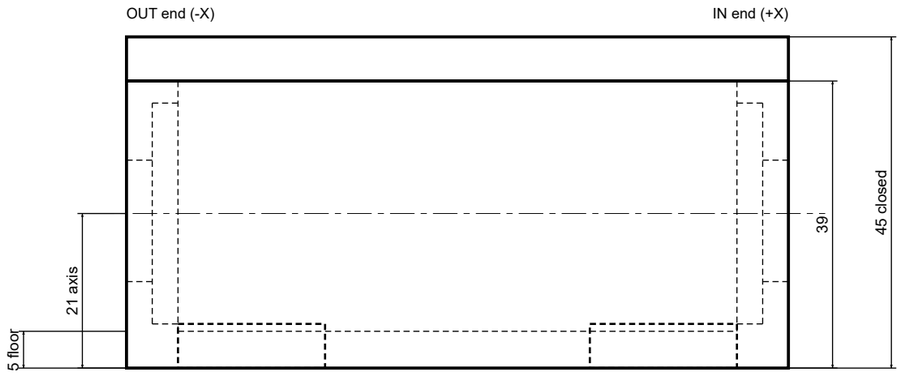
groove R 0.13-0.38, break edges 0.13, Ra 0.8 (32 rms) groove + lid face

squeeze 26 % nominal, fill 71 %; AS568-039 stretched ~2.4 % on centreline

LID 90 x 50 x 6 scale 1:1

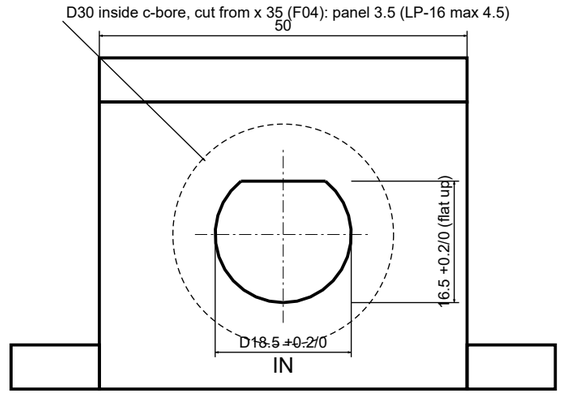


SIDE VIEW (from -Y), lid fitted scale 1.2:1



Tabs 6 thick (hidden). Cavity depth 34. Lid 6.

END VIEW +X (IN) scale 1.2:1



engrave IN / OUT Arial Black 4.0 high (strokes ~1.2), 0.3 deep, upright + readable from outside

NOTES - VRC box (x2 identical: front FL/FR, rear RL/RR)

1. 6061-T6 billet body + lid. Stock est.: body 95 x 80 x 42, lid 95 x 55 x 8 (plate).
2. Seal: AS568-039 O-ring (ID 69.57, W 1.78; Parker Table 9-1) in body-rim face groove. NBR 70, Parker N0674-70 2-039 (Dan E4). Lid closes metal-to-metal on the rim.
3. Lid: 4x M3 x 12 ISO 10642 A2 countersunk; tapped holes outside the O-ring.
4. Connectors: 2x CNLINKO LP-16-C05SX-02-101, D18.5/16.5 flat, M18x1.5 rear nut, 1.0 N.m, panel 3.5 (max 4.5) - CNLINKO drawing LP-16-xxxSX-02-1xxx rev A1. Behind panel 12.6 (derived: A 26.2 - front 13.6). Rear nut OD not published: D30 counterbore (inside face, as modelled) is a safe allowance - check a sample nut.
5. Electronics: OpenLogicEFI Dual VR Conditioner (DetonationEMS, Dan E2) held by a 3D-printed holder v2 on the box floor (Dan M9, R126): one PA12 block, board flush in a pocket with 4 snap tabs, wires soldered on the board top. 2x M6 x 12 ISO 10642 A2 csk (RS 266-8419).
6. Board holder: printed PA12, ST185_VRC_board_holder_PA12 (print/ .3mf / .stl / .step).
7. Mount on 4x M6 nylon screws (SHIELD-RULES: box mounts on nylon) - Dan E5.
8. BREAK SHARP EDGES 0.2-0.5 EXCEPT O-RING / SEAL FACES. No coating on rim / groove / lid seal face.
9. Inside D30 counterbores stay as modelled (Dan, 2026-10-14). 3D printed: as modelled. Machined: back-counterbore tool through the D-hole, or a second setup on that wall.
10. v2 holder (2026-10-16): the 4 M3 floor bosses of R108 are removed (suppressed in Fusion).
11. Tab-to-body inside corners R3 (F11). Floor-to-wall inside corners: CUTTER CORNER OK (square end mill).
12. Tab slots 6.6 x 10 (F01). Lid M3: drill 2.5 x 9.5 + 118 deg, thread 8 (F08).

ST185 VRC ENCLOSURE

Body + lid, O-ring face seal, LP-16 IN / OUT on opposite ends

Units mm | 3rd-angle | scales as noted | 6061-T6 | General tol. ISO 2768-m unless noted

Claude Code 2026-10-13, rev 2026-10-16 (SteveCAD audit v3) | Fusion ST185-VRC-Enclosure v6 | DRAFT for Dan review